Cauchy-characteristic matching for the Z4c evolution system: method

z4c-CMM verified research loop[1]

[1] *generated from the mission-2 knowledge ledger (`knowledge-database/paper_z4c-CCM/`); every equation and number below is traceable to a verifier output under `results/numerical/`*

(Dated: June 13, 2026)

I. INTRODUCTION

Cauchy–characteristic matching (CCM) couples a strongly hyperbolic interior (Cauchy) evolution to a characteristic evolution reaching future null infinity, so that the worldtube supplying characteristic boundary data simultaneously receives the incoming radiative content computed on the characteristic grid; the interface becomes transparent rather than approximately absorbing [1, 2]. The present paper formulates and verifies CCM for the Z4c formulation of the Einstein equations [3–5]: the constraint-damped conformal decomposition in production use across finite-difference numerical-relativity codes, here the AthenaK implementation [6]. The presentation follows the organization of Ref. [1]: the Cauchy system and its boundary sector (Sec. II), the characteristic system (Sec. III), the matching algorithm (Sec. IV), and numerical verification (Sec. V). Every equation is one of: (a) a machine-verified result of this program (dark blue; verification labels [Vn] and the ledger nodes N1–N16); (b) standard literature content, cited at the equation; (c) an explicitly open obligation (labels O-Nx-y). The derivation chain is summarized as a directed acyclic graph in Fig. 1.

II. THE Z4C CAUCHY SYSTEM AND ITS BOUNDARY SECTOR

Black equations are standard literature content, cited at the equation; dark blue equations are the machine-verified new Z4c-CCM content (verification labels [Vn], Table II; ledger nodes in section titles).

A. Boundary frame and Z4c variables

$$ds^2 = -\alpha^2\, dt^2 + \gamma_{ij}\,(dx^i + \beta^i dt)(dx^j + \beta^j dt), \qquad n^\mu = \alpha^{-1}\,(1,\, -\beta^i), \tag{1}$$

$$s_i = \frac{\partial_i r}{\sqrt{\gamma^{rr}}}, \qquad s^i = \frac{\gamma^{ir}}{\sqrt{\gamma^{rr}}}, \qquad m^A = \tfrac{1}{\sqrt{2}}\big(e_2 + i\, e_3\big)^A, \tag{2}$$

$$\mathring{l}^\mu = \tfrac{1}{\sqrt{2}}\,(n^\mu + s^\mu), \qquad \mathring{k}^\mu = \tfrac{1}{\sqrt{2}}\,(n^\mu - s^\mu) \qquad [5]. \tag{3}$$

Z4c variables [5]:

$$\chi = (\det\gamma)^{-1/3}, \quad \tilde\gamma_{ij} = \chi\,\gamma_{ij}, \quad K = \hat K + 2\Theta, \quad \tilde A_{ij} = \chi\Big(K_{ij} - \tfrac13 \gamma_{ij} K\Big), \tag{4}$$

$$\partial_t \chi = \tfrac23\,\chi\,\big[\alpha(\hat K + 2\Theta) - D_i\beta^i\big], \tag{5}$$

$$\partial_t \tilde\gamma_{ij} = -2\alpha \tilde A_{ij} + \beta^k \tilde\gamma_{ij,k} + 2\tilde\gamma_{k(i}\,\beta^k_{\ ,j)} - \tfrac23\,\tilde\gamma_{ij}\,\beta^k_{\ ,k} \qquad [5]. \tag{6}$$

B. The Z4c evolution system

The Z4c formulation [3, 4] evolves the conformal variables (4) together with $\hat K = K - 2\Theta$, the conformal trace-free curvature $\tilde A_{ij}$, the constraint pair (Θ, Z_i) — with Z_i carried by $\tilde\Gamma^i = \tilde\gamma^{jk}\tilde\Gamma^i_{jk} + 2\tilde\gamma^{ij} Z_j$ — and the gauge (α, β^i). In vacuum, with D_i the γ-covariant derivative and $[\cdot]^{\mathrm{TF}}$ the γ-trace-free part, the evolution equations are [3]

$$\partial_t \hat K = \beta^i \partial_i \hat K - D^i D_i \alpha + \alpha\Big[\tilde A_{ij}\tilde A^{ij} + \tfrac13\big(\hat K + 2\Theta\big)^2\Big] + \alpha\,\kappa_1\big(1 - \kappa_2\big)\,\Theta, \tag{7}$$

$$\partial_t \tilde{A}_{ij} = \beta^k\partial_k \tilde{A}_{ij} + 2\tilde{A}_{k(i}\partial_{j)}\beta^k - \tfrac{2}{3}\,\tilde{A}_{ij}\,\partial_k\beta^k + \chi\big[\alpha\,R_{ij} - D_iD_j\alpha\big]^{\rm TF} + \alpha\Big[\big(\hat{K}+2\Theta\big)\tilde{A}_{ij} - 2\,\tilde{A}_{ik}\tilde{A}^k{}_j\Big], \tag{8}$$

$$\partial_t\Theta = \beta^i\partial_i\Theta + \tfrac{\alpha}{2}\Big[\mathcal{R} + \tfrac{2}{3}\big(\hat{K}+2\Theta\big)^2 - \tilde{A}_{ij}\tilde{A}^{ij}\Big] - \alpha\,\kappa_1\big(2+\kappa_2\big)\,\Theta, \tag{9}$$

$$\begin{aligned}\partial_t\tilde{\Gamma}^i &= \beta^k\partial_k\tilde{\Gamma}^i - \tilde{\Gamma}^k\partial_k\beta^i + \tfrac{2}{3}\,\tilde{\Gamma}^i\partial_k\beta^k + \tilde{\gamma}^{jk}\partial_j\partial_k\beta^i + \tfrac{1}{3}\,\tilde{\gamma}^{ij}\partial_j\partial_k\beta^k \\ &\quad - 2\,\tilde{A}^{ij}\partial_j\alpha + 2\alpha\Big[\tilde{\Gamma}^i{}_{jk}\tilde{A}^{jk} - \tfrac{3}{2}\,\tilde{A}^{ij}\partial_j\ln\chi - \tfrac{1}{3}\,\tilde{\gamma}^{ij}\partial_j\big(2\hat{K}+\Theta\big) - \kappa_1\,\tilde{\gamma}^{ij}Z_j\Big],\end{aligned} \tag{10}$$

with $\mathcal{R}$ the Ricci scalar of γ_{ij} and (κ_1, κ_2) the constraint-damping parameters (production: $\kappa_1 = 0.02$, $\kappa_2 = 0$). The gauge is 1+log slicing and the gamma-driver shift [3, 6],

$$\partial_t\alpha = \beta^i\partial_i\alpha - \mu_L\,\alpha^2\,\hat{K}, \qquad \partial_t\beta^i = \mu_S\,\tilde{\Gamma}^i - \eta\,\beta^i + \beta^j\partial_j\beta^i, \tag{11}$$

(AthenaK realization: $\mu_L = 2/\alpha$, $\mu_S = 1$, $\eta = 2$ [6]). The constraints are

$$\mathcal{H} = \mathcal{R} + \tfrac{2}{3}K^2 - \tilde{A}_{ij}\tilde{A}^{ij}, \qquad \mathcal{M}_i = D^j\big(K_{ij} - \gamma_{ij}K\big), \qquad \Theta = 0, \qquad Z_i = 0, \tag{12}$$

and at the principal level the constraint subsystem propagates as waves, $\Box\Theta \simeq 0 \simeq \Box Z_i$ [5], the fact underlying the boundary rows of Sec. II F and the stability analysis of node N13.

C. The ten incoming boundary modes

At a timelike outer boundary with frame (1)–(3), the linearized Z4c system carries ten incoming characteristic fields (taxonomy verified at nodes N4/N5: transverse-traceless = physical; h_{ss} = gauge; h_{sA} = constraint-vector; tangential trace = constraint-scalar): The constraint and gauge rows are those of Ref. [5] adapted to the Z4c variable set; the

Table I. Incoming boundary modes of Z4c-CCM and their data sources (ledger node N6). Per-speed outgoing characteristic vectors $\mathring{X}^a = (\mathring{n}^a + \sqrt{v}\,\mathring{s}^a)/\sqrt{2}$ at sector speed v [5]; production order $L \geq 1$.

#	sector	incoming field	speed	boundary condition (datum)
1–2	physical	$U^-_{AB} = \mathrm{TT}_2[E + \epsilon(s)B]_{AB}$	1	$U^-_{AB} \hat{=} -(\psi_0'\,\bar{m}_A\bar{m}_B + \text{c.c.})$ [CCM: $\psi_0' = A^2\psi_0^{\rm CCE}$]
3	constraint	Θ	1	$(r^2\,\mathring{l}\cdot\partial)^{L+1}\,\Theta \hat{=} 0$
4	constraint	Z_s	1	$(r^2\,\mathring{l}\cdot\partial)^{L+1}\,Z_s \hat{=} 0$
5–6	constraint	Z_A	1	$(r^2\,\mathring{l}\cdot\partial)^{L+1}\,Z_A \hat{=} 0$
7	gauge	α	$\sqrt{\mu_L}$	$(r^2\,\mathring{m}\cdot\partial)^{L+1}\,\alpha \hat{=} h_\alpha$
8	gauge	β_s	$\sqrt{\nu_s}$	$(r^2\,\mathring{k}_{\nu_s}\cdot\partial)^{L+1}\,\beta_s \hat{=} h_s$
9–10	gauge	β_A	$\sqrt{\mu_S}$	$(r^2\,\mathring{j}\cdot\partial)^{L+1}\,\beta_A \hat{=} h_A$

physical rows 1–2 carry the CCM datum and constitute the new content (Secs. II D–II E, II F). With $\psi_0' \to 0$ the set reduces exactly to the absorbing constraint-preserving scheme of Ref. [5] [V6]. The frozen-coefficient LF/GKS analysis of the constraint rows (node N13) shows that every matched-falloff advection realization of rows 3–6 is stable and absorbs the (Θ, Z_s) pair at $1 - O\big((\sigma/2\omega r)^2\big)$, that falloff-mismatched sink variants self-generate $O(10\%)$ pair reflection, and that the production default (the stock matched realization) is the admissible optimum.

D. Tetrad correspondence and Type-II boost (node N2)

Outgoing null label u with $u|_{\partial\Omega} = t$: $\partial_\mu u = (1,\, u_r,\, 0,\, 0)$,

$$g^{tt} + 2g^{tr}u_r + g^{rr}u_r^2 = 0 \quad\Longrightarrow\quad u_r = \frac{-g^{tr} - \sqrt{(g^{tr})^2 - g^{tt}g^{rr}}}{g^{rr}}, \qquad u_r\big|_{\rm flat} = -1. \tag{13}$$

Characteristic outgoing null vector [1, 2]:

$$\hat{l}^{\,\mu} = -e^{2\hat{\beta}}\,g^{\mu\nu}\partial_\nu u. \tag{14}$$

For arbitrary $(\alpha, \beta^i, \gamma_{ij})$,

$$\mathring{l}^\mu = \frac{\sqrt{2}}{2}\,\big(\alpha - \gamma_{ij}\beta^i s^j\big)\,e^{-2\hat{\beta}}\,\hat{l}^{\,\mu} \qquad \text{[V1: exact at 12 rational points; GPU residual } 9.3\times10^{-16}\text{]}, \tag{15}$$

$$A = \big(\alpha - \gamma_{ij}\beta^i s^j\big)\, e^{-2\hat\beta}, \qquad \psi_0' = A^2\,\psi_0, \tag{16}$$

i.e. the Choice-2 boost of Ref. [1] holds verbatim on the Z4c frame (3): no evolution-system variable enters (15).

ψ_0 on the characteristic grid [1, 2]:

$$\psi_0 = \Big(\frac{r\partial_r\beta - 1}{4Kr}\Big)\Big[(1{+}K)\partial_r J - \frac{J^2\partial_r\bar J}{1{+}K}\Big] + \frac{J(1{+}K^2)\,\partial_r J\,\partial_r\bar J}{8K^3} + \frac{1}{8K}\Big[\frac{J^2\partial_r^2\bar J}{1{+}K} - (1{+}K)\partial_r^2 J\Big] - \frac{J\bar J^2(\partial_r J)^2 + J^3(\partial_r\bar J)^2}{16K^3}. \tag{17}$$

E. Physical-mode dictionary (node N3, exact)

Electric/magnetic Weyl decomposition w.r.t. n^μ (vacuum):

$$E_{\mu\nu} = C_{\mu\alpha\nu\beta}\,n^\alpha n^\beta, \qquad B_{\mu\nu} = \tfrac12\,\epsilon_{\mu\alpha}{}^{\gamma\delta}C_{\gamma\delta\nu\beta}\,n^\alpha n^\beta, \qquad \epsilon_{\mu\nu\rho\sigma} = \sqrt{-g}\,[\mu\nu\rho\sigma]. \tag{18}$$

EXACT identity (pure Weyl algebra, generic 10-component tensor; no perturbation, no field equations) [V2a]:

$$\big(E_{\mu\nu} + \epsilon_\mu{}^{\kappa\lambda}s_\kappa B_{\lambda\nu}\big)\,m^\mu m^\nu \;=\; -\,\psi_0, \qquad \big(E_{\mu\nu} - \epsilon_\mu{}^{\kappa\lambda}s_\kappa B_{\lambda\nu}\big)\,m^\mu m^\nu \;=\; -\,\bar\psi_4, \tag{19}$$

with $\psi_0 = -C_{abcd}\,l^a m^b l^c m^d$ on the frame (3); dyad completeness ($U_{AB} = U(m,m)\,\bar m_A\bar m_B + \text{c.c.}$ for tangential trace-free U) gives the tensor form of the incoming Weyl object

$$U^-_{AB} \;\equiv\; \mathrm{TT}_2\big[E + \epsilon(s)B\big]_{AB} \;=\; -\big(\psi_0\,\bar m_A\bar m_B + \bar\psi_0\,m_A m_B\big). \tag{20}$$

Realization from the Z4c state (Gauss–Codazzi, on-shell vacuum; γ_{ij}, K_{ij} from (4)) [V2c]:

$$E_{ij} = R_{ij}[\gamma] + K\,K_{ij} - K_{ik}K^k{}_j, \qquad B_{ij} = \epsilon_i{}^{kl}\,D_k K_{lj}\big|_{\mathrm{sym}}, \qquad \epsilon_{ikl} = \sqrt{\det\gamma}\,[ikl]. \tag{21}$$

Nonperturbative verification (Kerr–Schild Schwarzschild, autodiff, both routes; Table II V2c): $|E^{3+1}{-}E^{4D}|$ rel. $7.7{\times}10^{-17}$, $|B^{3+1}{-}B^{4D}|$ rel. $3.9{\times}10^{-17}$, identity residual $8.5{\times}10^{-16}$. Linearized on-shell flat limit (superseded defining form, now a corollary [V2b]): for TT $h_{AB}(t,x_s)$, $\psi_0 = \frac14(\partial_t+\partial_s)^2 h_{mm}$ and $U^-_{AB} \to -\frac14(\partial_t+\partial_s)^2\gamma^{\mathrm{TF}}_{AB}$, i.e. $(\partial_t+\partial_s)^2\gamma^{\mathrm{TF}}_{AB} = 4(\psi_0\bar m_A\bar m_B + \text{c.c.})$.

F. The Z4c-CCM boundary conditions (node N6)

Ten incoming modes at the worldtube; $\mathring{X}^\mu_v = (n^\mu + \sqrt{v}\,s^\mu)/\sqrt2$ at sector speed v; $L \ge 1$:

$$\begin{aligned}
&\text{physical (2):} && U^-_{AB} \mathrel{\hat=} -\Big(\psi_0'\,\bar m_A\bar m_B + \bar\psi_0'\,m_A m_B\Big), \qquad \psi_0' = A^2\,\psi_0^{\mathrm{CCE}} \quad (\text{exact, (20)–(21)}), && (22)\\
&\text{constraint (4):} && \big(r^2\,\mathring{l}\cdot\partial\big)^{L+1}\Theta \mathrel{\hat=} 0, \qquad \big(r^2\,\mathring{l}\cdot\partial\big)^{L+1}Z_s \mathrel{\hat=} 0, \qquad \big(r^2\,\mathring{l}\cdot\partial\big)^{L+1}Z_A \mathrel{\hat=} 0, && (23)\\
&\text{gauge (4):} && \big(r^2\,\mathring{m}\cdot\partial\big)^{L+1}\alpha \mathrel{\hat=} 0, \qquad \big(r^2\,\mathring{k}_{\nu_s}\cdot\partial\big)^{L+1}\beta_s \mathrel{\hat=} 0, \qquad \big(r^2\,\mathring{j}\cdot\partial\big)^{L+1}\beta_A \mathrel{\hat=} 0. && (24)
\end{aligned}$$

Bjørhus form of (22) (the correction drives the incoming Weyl object to its CCM value, the Z4c analog of Ref. [1]):

$$U^-_{AB} - U^-_{AB}\big|_{\mathrm{BC}} \to 0, \qquad U^-_{AB}\big|_{\mathrm{BC}} = -\Big(\psi_0'\,\bar m_A\bar m_B + \bar\psi_0'\,m_A m_B\Big); \qquad \psi_0^{\mathrm{CCE}} \to 0 \implies \text{CPBC scheme of Ref. [5]} \quad \text{[V6]}. \tag{25}$$

Equations (23)–(24) are the order-L conditions of Ref. [5]; (24) replaces the Sommerfeld gauge placeholder of Ref. [1].

G. Sector orthogonality (nodes N4, N5)

For arbitrary TT data $h_{AB}(t,x_s)$ [V4]:

$$\mathcal{H} = \partial_i\partial_j h_{ij} - \partial^2 h^k{}_k = 0, \qquad \mathcal{M}_i = \partial_j\dot h_{ij} - \partial_i\dot h^j{}_j = 0, \qquad \gamma^{ij}\partial_t h_{ij} = 0, \qquad \tilde\Gamma^i = -\partial_j\tilde h^{ij} = 0; \tag{26}$$

negative controls: $h_{x_s y} = G(t,x_s) \Rightarrow \mathcal{M}_y = -\frac{1}{2}\,\partial_{x_s}\dot{G} \neq 0$; $h_{yy} = h_{zz} = G \Rightarrow \mathcal{H} = -2\,\partial^2_{x_s} G \neq 0$.

Hence the datum (22) sources neither (23) nor (24) at linear order: the three sectors of (22)–(24) commute.

On the boundary wave model of Ref. [5], $\big[\partial_t^2 - 2\mathring{\beta}\,\partial_t\partial_x - (1-\mathring{\beta}^2)\partial_x^2\big]u = 0$, plane-wave reflection [V5]:

$$R_{\text{Sommerfeld}} = -\frac{\mathring{\beta}\,(1-\mathring{\beta})}{(1+\mathring{\beta})(2-\mathring{\beta})} \neq 0, \qquad R_{(\partial_t + c_+\partial_x)} = 0 \quad \text{(identically).} \tag{27}$$

H. Frozen-coefficient boundary stability (node N7)

Laplace–Fourier ansatz $u = \tilde{u}\,e^{st+\lambda x}$, $\Re s > 0$ [V7]:

$$\lambda_+ = \frac{s}{1+\mathring{\beta}} \text{ (admissible)}, \qquad \lambda_- = -\frac{s}{1-\mathring{\beta}}, \qquad D_L(s) = \Big(s + (1+\mathring{\beta})\,\lambda_+\Big)^{L+1} = (2s)^{L+1}, \qquad \frac{|D_L(s)|}{|s|^{L+1}} = 2^{L+1}. \tag{28}$$

D_L contains no datum symbol: the injection (22) modifies only the inhomogeneity, so the boundary-stability results of Ref. [5] (constraint sector: all L; gauge+metric sector: spherical reduction, asymptotically harmonic shift) transfer verbatim with $\|u\| \le C\,\|\psi_0'\text{-datum}\|$.

III. THE CHARACTERISTIC SYSTEM

A. Bondi–Sachs framework

The exterior of the worldtube $r = r_{\text{wt}}$ is evolved on outgoing null slices in Bondi–Sachs form [2, 7, 8],

$$ds^2 = -\Big(e^{2\beta}\tfrac{V}{r} - r^2 h_{AB}U^A U^B\Big)du^2 - 2e^{2\beta}\,du\,dr - 2r^2 h_{AB}U^B\,du\,dx^A + r^2 h_{AB}\,dx^A dx^B, \tag{29}$$

with the spin-weighted scalars built from the complex dyad q^A ($q^A q_A = 0$, $q^A\bar{q}_A = 2$): the radiative variable $J = \frac{1}{2}q^A q^B h_{AB}$, the lapse-like β, the shift-like $U = q_A U^A$ with its radial momentum $Q = r^2 e^{-2\beta} q^A h_{AB}\,\partial_r U^B$, the mass-like $W = (V-r)/r^2$, and the time derivative $H = \partial_u J$ at fixed compactified radius [2]. The radial coordinate is compactified as

$$y = 1 - \frac{2\,r_{\text{wt}}}{r}, \qquad y \in [-1,1], \qquad \frac{dy}{dr} = \frac{(1-y)^2}{2\,r_{\text{wt}}}, \tag{30}$$

so the worldtube sits at $y=-1$ and future null infinity at $y=1$. On each null slice the Einstein equations order into the radial hypersurface hierarchy $\beta \to Q \to U \to W \to H$ (each equation an ODE in r sourced by the previous members and J), followed by the evolution equation $\partial_u J = H$ [1, 2, 7]; we do not reproduce the nonlinear system here and work instead with its linearization, which the program verifies term by term.

B. Linear $\ell = 2$ scalarization

At linear order about Minkowski, the axisymmetric $(\ell, m) = (2, 0)$ sector reduces to one real scalar per field [machine-derived and verified, ledger node N14]: with $s \equiv \sin\theta$, $c \equiv \cos\theta$, $P \equiv 3c^2 - 1$,

$$J = J_t\,s^2, \qquad Q = Q_t\,sc, \qquad U = U_t\,sc, \qquad W = W_t\,P, \qquad \beta = \beta_t\,P, \qquad H = H_t\,s^2, \tag{31}$$

and the $\ell = 2$ spin-weighted derivatives close on the same shapes,

$$\eth\beta = -6\,\beta_t\,sc, \qquad \bar{\eth}\eth\beta = -6\,\beta_t\,P, \qquad \eth^2\beta = 6\,\beta_t\,s^2, \qquad \bar{\eth}Q = -Q_t\,s^2, \qquad \bar{\eth}U = -U_t\,s^2. \tag{32}$$

The spin-weighted spherical-harmonic modal coefficients used for code interoperation follow from

$$s^2 = 4\sqrt{\tfrac{2\pi}{15}}\;{}_2Y_{20}, \qquad sc = -\sqrt{\tfrac{8\pi}{15}}\;{}_1Y_{20}, \qquad P = \sqrt{\tfrac{16\pi}{5}}\;Y_{20}. \tag{33}$$

In the Bondi gauge ($\beta = 0$) the verified linear hierarchy is

$$\partial_r\big(r^2 Q_t\big) = -4\,r^2\,\partial_r J_t, \qquad\qquad \partial_r U_t = \frac{Q_t}{r^2},$$

$$\partial_r\left(r^2W_t\right) = 2J_t + 2rU_t + \frac{r^2}{2}\partial_r U_t, \qquad \partial_r\left(rH_t\right) = -\frac{U_t}{2} + \frac{r}{4}\,\partial_r^2 J_t - \frac{3J_t}{2r}, \tag{34}$$

where the H row is the plain-family pinned combination; the joint exact-rational refit on the enlarged (gauge-shifted) solution family fixes the equivalent covariant form (35) below [derivation artifact `derive_beta_hierarchy.jl`].

C. The β-corrected hierarchy for live (anchored-gauge) data

Worldtube data constructed from a live Cauchy state arrive in the worldtube-anchored gauge of Sec. IV, which carries $\beta \neq 0$ (r-independent at this order) and non-decaying gauge tails. The hierarchy sources required by such data were machine-derived as exact-rational fits on the anchored and plain solution families jointly:

$$\begin{aligned}
&\partial_r(r^2Q) = -\,r^2\,\partial_r(\bar{\eth}J) - 4r\,\eth\beta, && \partial_r U = Q/r^2,\\
&\partial_r(r^2W) = \tfrac{1}{2}\bar{\eth}\bar{\eth}J + 2r\,\bar{\eth}U + \tfrac{r^2}{2}\partial_r(\bar{\eth}U) - \tfrac{4}{3}\,\bar{\eth}\eth\beta, && \partial_r(rH) = \tfrac{1}{2}r\,\partial_r^2 J + \partial_r J + \frac{\eth^2\beta}{r} - \frac{\eth Q}{2r} - \eth U.
\end{aligned} \tag{35}$$

Scalarized with (31)–(32) and recast on the compactified coordinate (30), the system becomes four ODEs with coefficients regular on the whole interval $y \in [-1,1]$ in *both* gauges (the $y=1$ rows are automatic regularity conditions):

$$\begin{aligned}
&2Q_t + (1{-}y)\,Q_t' = -4(1{-}y)\,J_t' + 24\,\beta_t, \qquad U_t' = \frac{Q_t}{2r_{\rm wt}},\\
&2W_t + (1{-}y)\,W_t' = \frac{(1{-}y)J_t}{r_{\rm wt}} + 2U_t + \frac{(1{-}y)U_t'}{2} + \frac{4\beta_t(1{-}y)}{r_{\rm wt}},\\
&H_t + (1{-}y)\,H_t' = \frac{(1{-}y)}{4r_{\rm wt}}\Big[(1{-}y)^2J_t'' - 2(1{-}y)J_t'\Big] + \frac{(1{-}y)^2}{2r_{\rm wt}}J_t' + \frac{3\beta_t(1{-}y)}{r_{\rm wt}} + \frac{Q_t(1{-}y)}{4r_{\rm wt}} + U_t,
\end{aligned} \tag{36}$$

with $' = d/dy$. The *ringed* variables $rJ_t,\ r^2U_t,\ \ldots$ diverge at $y=1$ on anchored-gauge data (the $k=0,1$ tail monomials); the unringed form (36) is the representation in which the anchored solution is finite everywhere [measured failure of the ringed scheme on anchored data: ψ_0 off by $O(10^{3\text{–}4})$; pinned as a regression test].

D. Spectral solver, ψ_0 extraction, and pulse data

The radial direction is discretized on Chebyshev–Gauss–Lobatto nodes with the standard differentiation matrix; the linear solves of (36) use LU factorizations built once, each operator's boundary row imposing the worldtube value. Time advance is RK4 on $\partial_u J_t = H_t$ with a penalty (SAT) imposition of the worldtube Dirichlet value,

$$H_t\big|_{y=-1} \mathrel{+}= \tau\left(J_t^{\rm wt} - J_t|_{y=-1}\right), \qquad \tau = \frac{\sigma N^2}{2\,r_{\rm wt}}, \quad \sigma = 1. \tag{37}$$

The incoming Weyl datum is extracted from J_t alone,

$$\psi_0^t(r) = -\frac{\partial_r J_t}{2r} - \frac{\partial_r^2 J_t}{4}, \qquad \psi_0 = \psi_0^t\,s^2, \tag{38}$$

an expression *gauge-invariant on the anchored family*: the anchored tails $a(u) + b(u)/r$ satisfy $\psi_0^t[a + b/r] = 0$ identically, and the machine check $\psi_0[J_{\rm anchored}] = \psi_0[J_{\rm Bondi}]$ holds as an exact rational identity [N14]. Evaluation at an interior radius $r_* \geq r_{\rm wt}$ (required by the matching geometry of Sec. IV) uses barycentric interpolation at $y_* = 1 - 2r_{\rm wt}/r_*$ together with

$$\partial_r J_t = \frac{(1{-}y)^2}{2r_{\rm wt}}\,J_t', \qquad \partial_r^2 J_t = \frac{(1{-}y)^2}{4r_{\rm wt}^2}\Big[(1{-}y)^2J_t'' - 2(1{-}y)J_t'\Big]; \tag{39}$$

the interior point is *better* conditioned than the worldtube node (measured: evolved-probe relative error 8.8×10^{-5} vs the $\sim 6\times10^{-2}$ worldtube floor at the same resolution and step). For the pulse-injection test of Sec. V the initial null slice carries the compactly supported $\ell = 2$ free datum specified in the test design of Ref. [1] (polynomial envelope times Gaussian in y, amplitude Z, support $[y_{\min}, y_{\max}]$, center y_c, width τ_y), converted to the t-scalar convention by

$$J_t(y)\big|_{u_0} = \frac{1}{4}\sqrt{\frac{15}{2\pi}}\;\mathcal{J}(y), \qquad \mathcal{J}(y) = \frac{4Z\,(y_{\max} - y)(y - y_{\min})}{(y_{\max} - y_{\min})^2}\,e^{-(y-y_c)^2/\tau_y^2} \ \text{ on } [y_{\min}, y_{\max}], \tag{40}$$

and zero outside the support.

E. Exact Teukolsky reference in Bondi variables

The solver verifier is the exact linear Teukolsky solution (Sec. V B, Eqs. (51)–(54)) transformed to the Bondi gauge; the machine-derived closed forms are

$$
\begin{aligned}
J &= \tfrac{3}{4}\, s^2 \Big[\frac{F^{(4)}}{r} + \frac{F^{(2)}}{r^3}\Big], \qquad \beta = 0, \qquad H = \partial_u J,\\
Q &= sc\Big[\frac{3F^{(4)}}{r} - \frac{9F^{(3)}}{r^2} - \frac{9F^{(2)}}{r^3}\Big], \qquad U = sc\Big[-\frac{3F^{(4)}}{2r^2} + \frac{3F^{(3)}}{r^3} + \frac{9F^{(2)}}{4r^4}\Big],\\
W &= P\Big[-\frac{3F^{(4)}}{2r^2} - \frac{3F^{(3)}}{2r^3} - \frac{3F^{(2)}}{4r^4}\Big],
\end{aligned} \tag{41}
$$

all arguments retarded, $F^{(n)} = F^{(n)}(u)$ with $u = t - r$. The hierarchy (34) closes on (41) as a rational identity; the incoming datum is

$$\psi_0 = -\tfrac{9}{8}\, s^2\, \frac{F^{(2)}(u)}{r^5} \qquad \text{(exactly; no subleading retarded terms)}, \tag{42}$$

consistent with (56), and the outgoing waveform is the finite-radius expression (59) with the convention map stated there. An independent production characteristic code run on worldtube data built from (41) reproduces $\Psi_4 = -\sqrt{6\pi/5}\, F^{(6)}$ at future null infinity to 6×10^{-6} relative and (42) to 0.1% (Sec. V).

F. Solver verification summary

(i) Exact-slice sweeps of (36) are machine-exact ($\sim 10^{-22}$) on both the Bondi and anchored families. (ii) The SAT evolution reproduces the worldtube ψ_0 at the documented double-precision conditioning floor and the interior probe (39) at 8.8×10^{-5} relative (7.1×10^{-6} at halved step; truncation-dominated), identically for plain and anchored boundary data. (iii) The native C++ production port is gated against the verification-harness numbers (standalone test; relative agreement $\sim 10^{-7}$). (iv) The pulse problem (40) is radially converged at $n = 65$ collocation points to 2.1×10^{-7} of peak (resolution ladder), with time-stepping error 4×10^{-12}. (v) The constraint-row boundary analysis and the live two-way coupling are treated in Secs. II and IV–V respectively.

IV. THE MATCHING ALGORITHM

The matching loop closes two data paths each Cauchy step: the worldtube path (Cauchy state → characteristic boundary data, Secs. IV A–IV C) and the datum path (characteristic ψ_0 → the physical boundary rows of Sec. II F, Secs. IV D–IV E). The full derivation chain is summarized in Fig. 1.

A. Worldtube data map (node N1)

$$g_{00} = -\alpha^2 + \gamma_{ij}\beta^i\beta^j, \qquad g_{0i} = \gamma_{ij}\beta^j, \qquad g_{ij} = \gamma_{ij}; \qquad g^{00} = -\alpha^{-2}, \quad g^{0i} = \frac{\beta^i}{\alpha^2}, \quad g^{ij} = \gamma^{ij} - \frac{\beta^i\beta^j}{\alpha^2}. \tag{43}$$

ADM kinematics requires $\partial_t\gamma_{ij} = -2\alpha K_{ij} + \beta^k\partial_k\gamma_{ij} + \gamma_{kj}\partial_i\beta^k + \gamma_{ik}\partial_j\beta^k$. From $\gamma_{ij} = \chi^{-1}\tilde\gamma_{ij}$,

$$\partial_t\gamma_{ij} = \chi^{-1}\partial_t\tilde\gamma_{ij} - \chi^{-2}\tilde\gamma_{ij}\,\partial_t\chi, \tag{44}$$

and inserting (5)–(6) with $K_{ij} = \chi^{-1}\big(\tilde A_{ij} + \tfrac{1}{3}\tilde\gamma_{ij}K\big)$, $\beta^k\partial_k\gamma_{ij} = \chi^{-1}\beta^k\partial_k\tilde\gamma_{ij} - \gamma_{ij}\,\beta^k\partial_k \ln\chi$,

$$\big[\partial_t\gamma_{ij}\big]_{\mathrm{Z4c}} - \big[\partial_t\gamma_{ij}\big]_{\mathrm{ADM}} = \gamma_{ij}\Big[\tfrac{2}{3}\, D_k\beta^k - \tfrac{2}{3}\,\partial_k\beta^k + \beta^k\partial_k \ln\chi\Big]. \tag{45}$$

$$\boxed{D_i\beta^i = \partial_k\beta^k - \tfrac{3}{2}\,\beta^k\,\partial_k \ln\chi} \qquad [\text{V3: closure exact; naive } D = \partial_k\beta^k \text{ leaves residual } +1\cdot\gamma_{ij}\beta^k\partial_k \ln\chi]. \tag{46}$$

B. Live sampling and the $\ell = 2$ data contract

The Cauchy ADM state is interpolated to three geodesic spheres at $r_{\rm wt}$ and $r_{\rm wt} \pm \delta r$ each step. With orthonormal frame components $h_{\hat a\hat b} = (\gamma - \delta)_{\hat a\hat b}$ and $\partial_t h_{\hat a\hat b} = -2K_{\hat a\hat b}$ (linear order; $\alpha = 1 + O(h)$, the correction entering beyond linear), the $(\ell, m) = (2, 0)$ contract scalars are the shape projections

$$h_{\rm TT} = \Big\langle \tfrac{1}{2}(h_{\hat\theta\hat\theta} - h_{\hat\phi\hat\phi}) \Big\rangle_{s^2}, \quad h_{rr} = \langle h_{\hat r\hat r} \rangle_P, \quad h_{\rm tr} = \langle h_{\hat\theta\hat\theta} + h_{\hat\phi\hat\phi} \rangle_P, \quad h_{r\theta} = \langle r\, h_{\hat r\hat\theta} \rangle_{sc}, \tag{47}$$

with $\langle f \rangle_X = \int f X\, d\Omega / \int X^2\, d\Omega$ for $X \in \{s^2, P, sc\}$, the time derivatives obtained from the same projections of $-2K$, and $\partial_r h_{\rm tr}$ by the centered difference across the sphere triplet. At $\ell = 2$ the angular derivatives close analytically on the same scalars,

$$\big\langle \partial_\theta h_{\rm tr} \big\rangle_{sc} = -6\, h_{\rm tr}, \qquad \big\langle \partial_\theta \partial_r h_{\rm tr} \big\rangle_{sc} = -6\, \partial_r h_{\rm tr}, \qquad \big\langle (\partial_\theta + \cot\theta)\, h_{r\theta} \big\rangle_P = h_{r\theta}, \tag{48}$$

so the contract is seven numbers per step: $(h_{\rm TT}, \partial_t h_{\rm TT}, h_{rr}, h_{\rm tr}, \partial_r h_{\rm tr}, h_{r\theta}, \partial_t h_{r\theta})$.

C. The anchored worldtube map (six rows $+\beta$)

The Cauchy coordinates near the worldtube do not satisfy the Bondi gauge conditions; the radial gauge generator is fixed by areal-radius consistency only up to functions of (u, θ). Anchoring the generator to vanish *at the worldtube*, with the angular generator corrected so that the $g_{r\theta}$ gauge condition holds consistently ($\xi_a = \xi_{\rm old} + \partial_\theta(\zeta_u^{\rm wt})/r - \text{const}$), makes the map from the contract (47)–(48) to the characteristic boundary scalars *local*:

$$\begin{aligned}
J_t^{\rm wt} &= h_{\rm TT}, & H_t^{\rm wt} &= \partial_t h_{\rm TT}, & U_t^{\rm wt} &= \frac{6\, h_{\rm tr}}{4\, r_{\rm wt}}, \\
Q_t^{\rm wt} &= \frac{6\, h_{\rm tr}}{4} + \frac{6\, r_{\rm wt}\, \partial_r h_{\rm tr}}{4} - \frac{3\, h_{r\theta}}{r_{\rm wt}} + \partial_t h_{r\theta}, & W_t^{\rm wt} &= \frac{3\, h_{rr}}{4\, r_{\rm wt}} - \frac{\partial_r h_{\rm tr}}{4}, & \beta_t^{\rm wt} &= \frac{h_{rr}}{4} - \frac{h_{\rm tr}}{8} - \frac{r_{\rm wt}\, \partial_r h_{\rm tr}}{8} + \frac{h_{r\theta}}{4\, r_{\rm wt}},
\end{aligned} \tag{49}$$

verified as an exact identity against the anchored-gauge solution series (BigFloat-192 residual $< 10^{-35}$; derivation artifact `derive_wtmap_v2.jl`). The anchored gauge necessarily carries $\beta_t \neq 0$ and non-decaying tails $a(u) + b(u)/r$ in J, U, W; these are exactly the structures supported by the β-corrected, unringed system (36), and they are ψ_0-silent by (38). Independent anchoring of the radial and angular generators violates the $g_{r\theta}$ condition and is excluded [pinned: the uncorrected map fails the β channel with a characteristic $(4 - n)$-ratio pattern].

D. Retarded matching geometry

The characteristic time is the retarded cone label, $u = t - r$ on the flat background: the cone u meets the worldtube worldline at Cauchy time $t = u + r_{\rm wt}$ and the Cauchy outer boundary r_B at $t = u + r_B$. Hence

$$\text{tube sample at } t \;\longrightarrow\; \text{cone } u = t - r_{\rm wt}, \qquad \text{boundary datum at } t \;\longleftarrow\; \psi_0^t(r_B)\Big|_{u = t - r_B}, \tag{50}$$

a causality lag of $r_B - r_{\rm wt}$ in u that is always available in lockstep evolution. The solver records the interior probe (39) at r_B each substep and serves the lagged value by linear interpolation in the recorded history, with the analytic initial-data value as the pre-cone fallback (the run starts on the quiescent cone $u_0 = -r_{\rm wt}$). Boundary data are held fixed across the substeps between Cauchy samples — an $O(\Delta t)$ coupling approximation. Measured against the published reference data of Ref. [1] (Sec. V), this approximation, together with the tube-inside-datum geometry, produces $O(30\%)$ deviations of the live boundary ψ_0 from the clean characteristic solution in the pulse-arrival window; linear-in-time boundary-data interpolation is the corresponding open refinement [O-M5-1, class (c)].

E. Injection path, cadence, and initialization

The served datum enters the Cauchy evolution through the Type-II boost and the physical rows of Sec. II F: $\psi_0' = A^2\, \psi_0$ with $A = (\alpha - \gamma_{ij}\beta^i s^j)\, e^{-2\hat\beta}$ (16), injected as (58). Per Cauchy RK stage at the boundary: (1) sample the contract (47) (collective over ranks; off-rank sphere points are excluded by ownership, the shape norms being global); (2) map by (49); (3) advance the characteristic state to $u = t - r_{\rm wt}$ (root rank); (4) serve the lagged probe (50) and broadcast; (5) apply the boundary rows. The characteristic initial slice is the analytic datum (quiescent for the matched tests; (40) for the pulse-injection test); initialization order and the $\hat{\beta}$ circularity at $t = 0$ remain documented obligations [O-N6-3]. Setting the served datum to zero reduces the entire loop to the absorbing scheme of Ref. [5] exactly [V6].

F. Derivation chain

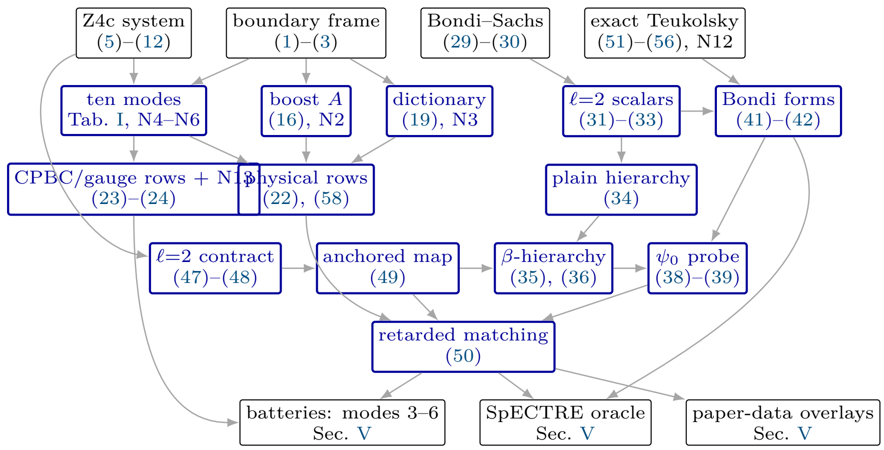

Figure 1. Derivation chain of the Z4c-CCM formulation. Black nodes: foundations (literature content and the exact reference chain); blue nodes: machine-verified new content (ledger nodes named in the section titles); bottom row: the verification instruments of Sec. V.

V. NUMERICAL VERIFICATION

A. Numerical evidence

All runs: JAX float64, 4× NVIDIA A100-80GB (`CUDA_VISIBLE_DEVICES=0,1,2,3`), wall clock ≤ 23 s each (budget 600 s); outputs under `results/numerical/`.

Table II. Symbolic and sampling verification of (46)–(28). "Exact" = sympy identity on arbitrary functions or exact rationals (zero float error). Files: `n2_boost_check.txt`, `n3_exact_check.txt`, `n3_dictionary_check.txt`, `n1_varmap_check.txt`, `n4_cpbc_compat_check.txt`, `n5_gauge_check.txt`, `n6_composite_check.txt`, `n7_lf_sketch_check.txt`.

	verified statement	exact part	GPU sweep (4 devices)
V1	boost identity (15), factor $\sqrt{2}/2$	12/12 rational points	9.34×10^{-16} over 4,194,304 samples
V2a	EXACT identity (19)–(20), $(\sigma, \kappa) = (1, -1)$, $\bar{\psi}_4$ partner	generic Weyl, all 10 dof	—
V2b	linearized on-shell limit (corollary)	arbitrary F_1, F_2	0.0 over 4,194,304 modes (prior form)
V2c	Gauss–Codazzi realization (21), Schwarzschild	Codazzi sign +1 derived	E: $7.7{\times}10^{-17}$, B: $3.9{\times}10^{-17}$, identity 8.5
V3	maps (43), closure (46)	full-symbol identities	round trip $4.10{\times}10^{-16}$, gg^{-1} $8.88{\times}10^{-16}$
V4	orthogonality (26) + controls	identities and non-vanishing controls	TT residual 0.0 vs control 12.3
V5	reflection (27)	exact R algebra	$\lvert R_{\mathrm{Som}} \rvert \geq 0.023$ on $\mathring{\beta} \in [0.05, 0.6]$; $R = 0$ ex
V6	10-mode count, $\psi_0 \to 0$ reduction, wire-through (exact targets)	structural	reduction 0.0; chain residual $1.47{\times}10^{-16}$
V7	Kreiss condition (28), $L = 0..3$	$D_L = (2s)^{L+1}$ exact	ratio deviation 3.2×10^{-14} (4,194,304 sam

B. Dynamical test at the parameters of Ref. [1] (nodes N10–N11)

Full 3+1 evolution: AthenaK Z4c [6] (vendored, pinned; RK4, CFL 0.25, sixth-order stencils at `nghost` $= 4$, $\kappa_1 = 0.02$), with (22) realized at the Cauchy outer boundary as described below. This subsection is self-contained: every symbol used by the test is defined here or in Secs. II A–II F.

Table III. Dynamical model evolution (6th-order interior stencils with a GKS-stable mixed closure — 4th-order Fornberg rows at three edge nodes; fully one-sided 6th-order closures are RK4-unstable (error ledger) — RK4, CFL 0.2, shift $\mathring{\beta} = 0.3$, outgoing Gaussian pulse onto the boundary; one configuration per GPU; `n8_model1d_test.txt`). Reflection is measured in the L_2 (energy) norm — trapezoid sums of Gaussian pulses converge exponentially, so the deviation column is limited by machine arithmetic, not by measurement or truncation. Analytic L_2 relation: $\|q_{\rm in}\|_{L_2}/\|q_{\rm out}\|_{L_2} = \frac{\mathring{\beta}}{2+\mathring{\beta}}\sqrt{\frac{1+\mathring{\beta}}{1-\mathring{\beta}}}$. Each row sits at its float64 floor: rows 1–3 at the per-sample roundoff floor (deviation saturates between N=8193 and N=16385: 1.9 vs 2.1×10^{-14}); the injection row at the linear roundoff-accumulation optimum ($\sim$ steps $\times\, \varepsilon$; ladder N=16385/32769/65537/131073 $\Rightarrow$ 5.4/2.4/5.9/30 $\times 10^{-11}$, non-monotone and insensitive to CFL 0.2 vs 0.05).

boundary condition	measured	exact	deviation
Sommerfeld, L_2 ratio (N=16385)	0.177752646227	0.177752646227	2.1×10^{-14} (float64 floor)
Sommerfeld, L_2 ratio (N=8193)	—	—	1.9×10^{-14} (saturated)
$(\partial_t + c_+\partial_x)$ [P1/CCM op., N=16385]	$R_{L_2} = 1.9 \times 10^{-14}$	0	machine precision
CCM injection (25), rel. L_2 (N=32769)	2.4×10^{-11}	0	roundoff-accumulation optimum

a. Exact solution injected. Even-parity $(\ell, m) = (2, 0)$ Teukolsky wave [9] with the Gaussian profile of Ref. [1], written with physicists' Hermite polynomials H_n and the origin-regular retarded/advanced combinations:

$$F(u) = X\, e^{-(u-r_c)^2/\tau^2}, \qquad F^{(n)}(u) = X\left(-\frac{1}{\tau}\right)^n H_n\!\left(\frac{u-r_c}{\tau}\right) e^{-(u-r_c)^2/\tau^2}, \tag{51}$$

$$Q^{(n)}(t,r) = F^{(n)}(t-r) - (-1)^n F^{(n)}(t+r), \qquad R^{(n)}(t,r) = F^{(n)}(t-r) + (-1)^n F^{(n)}(t+r), \qquad \partial_t Q^{(n)} = R^{(n+1)}. \tag{52}$$

Radial functions (regular at $r = 0$ by (52); calligraphic to keep A for the boost (16)):

$$\mathcal{A} = 3\left[\frac{Q^{(2)}}{r^3} + \frac{3Q^{(1)}}{r^4} + \frac{3Q^{(0)}}{r^5}\right], \qquad \mathcal{B} = -\left[\frac{Q^{(3)}}{r^2} + \frac{3Q^{(2)}}{r^3} + \frac{6Q^{(1)}}{r^4} + \frac{6Q^{(0)}}{r^5}\right],$$
$$\mathcal{C} = \frac{1}{4}\left[\frac{Q^{(4)}}{r} + \frac{2Q^{(3)}}{r^2} + \frac{9Q^{(2)}}{r^3} + \frac{21Q^{(1)}}{r^4} + \frac{21Q^{(0)}}{r^5}\right], \tag{53}$$

orthonormal-frame metric perturbation ($\dot{h}_{ij}$ from $Q^{(n)} \to R^{(n+1)}$ termwise):

$$h_{\hat r\hat r} = \mathcal{A}\,\frac{1+3\cos 2\theta}{2}, \quad h_{\hat r\hat\theta} = -3\mathcal{B}\sin\theta\cos\theta, \quad h_{\hat\theta\hat\theta} = 3\mathcal{C}\sin^2\theta - \mathcal{A}, \quad h_{\hat\phi\hat\phi} = -3\mathcal{C}\sin^2\theta + \mathcal{A}\,(3\sin^2\theta - 1), \tag{54}$$

$$\gamma_{ij}(0,x) = \delta_{ij} + h_{ij}(0,x), \qquad K_{ij}(0,x) = -\tfrac{1}{2}\,\dot{h}_{ij}(0,x) \qquad \text{(linear order; Ref. [1] solves XCTS).} \tag{55}$$

Weyl scalars of the solution at linear order (the self-datum and the extraction reference):

$$r^5\,\psi_0(t,r,\theta) = -\sqrt{\tfrac{27\pi}{10}}\;F^{(2)}(t-r)\;{}_2Y_{20}(\theta), \qquad {}_2Y_{20} = \tfrac{1}{4}\sqrt{\tfrac{15}{2\pi}}\,\sin^2\theta, \qquad r\,\psi_4\big|_{(2,0)} \;\to\; -\sqrt{\tfrac{6\pi}{5}}\,F^{(6)}(t-r). \tag{56}$$

b. Discrete realization of (22). At every Cauchy-boundary point, after the CPBC/gauge rows (23)–(24) (Sommerfeld task of Ref. [6]), the two physical rows are driven through $\tilde{A}_{ij}$:

$$s^a = \frac{x^a/r}{\sqrt{\gamma_{bc}\,x^b x^c}/r}, \qquad e_\phi \propto \hat{z}\times s, \quad e_\theta = s\times e_\phi, \qquad \bar{m} = \tfrac{1}{\sqrt{2}}\,(e_\theta - i\,e_\phi), \tag{57}$$

$$\boxed{\partial_t \tilde{A}_{ab}\big|_{\partial\Omega} \mathrel{+}= -\frac{2\chi}{\alpha}\left(\psi_0'\,\bar{m}_a\bar{m}_b + \bar{\psi}_0'\,m_a m_b\right), \qquad \psi_0' = A^2\,\psi_0^{\rm datum}, \quad A = \left(\alpha - \gamma_{ij}\beta^i s^j\right)e^{-2\hat\beta}} \tag{58}$$

with $\psi_0^{\rm datum}$ the right-hand member of (56) evaluated at the boundary point ($\hat\beta = 0$ on the flat background); $\psi_0^{\rm datum} \to 0$ reduces (58) to the unmodified scheme exactly (V6; machine-zero reduction in Table III methodology, AthenaK battery: Minkowski preserved to 2.6×10^{-33}).

c. Exact finite-radius reference (node N12). The extraction reference is the *analytic* waveform at the extraction radius itself (no auxiliary reference run): in the AthenaK extraction convention ($= -1\times$ the scri member of (56) continued to finite r),

$$r\,\psi_4^{(2,0)}(t,r) = \sqrt{\tfrac{6\pi}{5}}\left[F^{(6)} + \frac{2F^{(5)}}{r} + \frac{3F^{(4)}}{r^2} + \frac{3F^{(3)}}{r^3} + \frac{3F^{(2)}}{2\,r^4}\right](t-r) \;-\; \sqrt{\tfrac{6\pi}{5}}\,\frac{3F^{(2)}}{2\,r^4}(t+r), \tag{59}$$

verified by seven symbolic gates (tracelessness, coefficient-exact linearized vacuum, $\sin^2\theta$ factorization, peeling, scri limit, independent 30-digit finite-difference cross-check at 10^{-40}) and — independently — by the SpECTRE `CharacteristicExtract` cross oracle below.

d. Protocol and result. Parameters of Ref. [1]: $r_c = 20$, $\tau = 2$, $X = 10^{-5}$, Cauchy boundary (worldtube analog) at 41, $t \in [0, 80]$; extraction of $r\psi_4^{(2,0)}$ at $r = 36$; resolution ladder $h = 0.5,\ 0.331,\ 0.277$ (4×A100, every run < 600 s). Error against (59) at each rung's own output times:

$$E(h) = \frac{\left\| r\psi_4^{(2,0)}[h] - r\psi_4^{(2,0)}[\text{exact}] \right\|_2}{\left\| r\psi_4^{(2,0)}[\text{exact}] \right\|_2} = 4.99 \times 10^{-2},\quad 5.36 \times 10^{-3},\quad 2.02 \times 10^{-3}, \tag{60}$$

with measured convergence order 5.39 (first pair) and 5.52 (second pair), finest-rung amplitude $1 \pm 2 \times 10^{-4}$ and time shift 0.000 (Fig. 2). Two production defects were found *only* by this exact-reference program and are corrected in all data shown: a frame-orientation error in the initial data ($\hat{e}_\theta$ flipped; linearized vacuum violated at 5% of $|R_{ij}|$, invisible to constraint monitors and to run-vs-reference protocols since it sat on both sides) and a one-step offset in the waveform time labels. The self-datum CCM−Sommerfeld difference remains peeling-suppressed ($\psi_0 \sim r^{-5}$, relative imprint $\sim 10^{-9}$ of peak at this worldtube), as in Ref. [1] Sec. V.A.

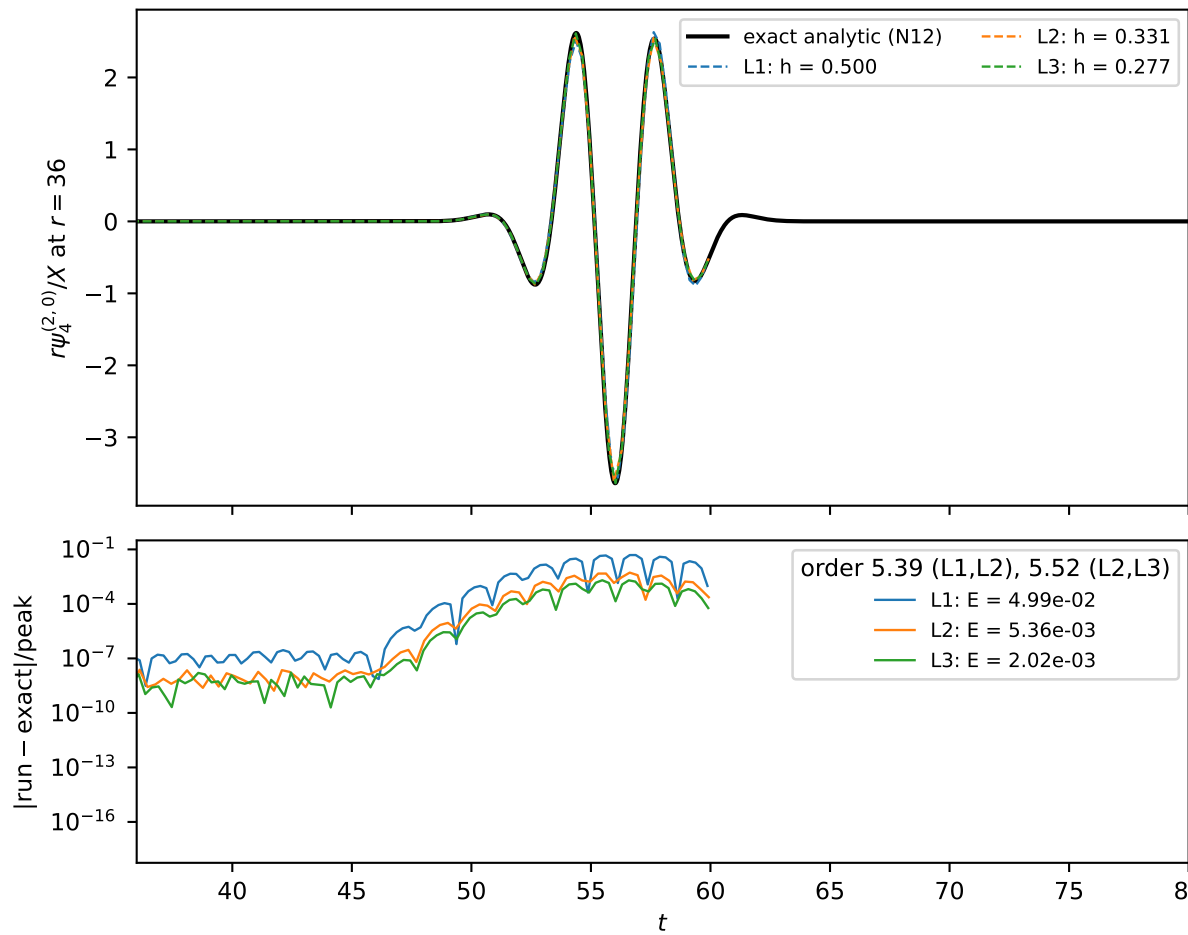

Figure 2. Exact-reference Teukolsky test of Sec. V B (r_c=20, τ=2, boundary 41, extraction r=36, X=10^{-5}). Top: (ℓ, m)=(2, 0) mode of $r\psi_4$ — the exact finite-radius waveform (59) (solid) and the three-rung sixth-order ladder. Bottom: pointwise errors against the exact reference; measured order 5.39/5.52, finest-rung relative L_2 error 2.02×10^{-3} (60). Generator: `scripts/plot_paper_figures_v2.py`; data: `results/numerical/athenak_teuk_ana/gpu6/` (ledger node N12).

C. Reproduction against published CCM simulation data

We verify the formulation against the published simulation data of Ref. [1] (the authors' archived dataset and figure scripts), regenerating their figures from their data and overlaying our independent pipeline. All twenty data

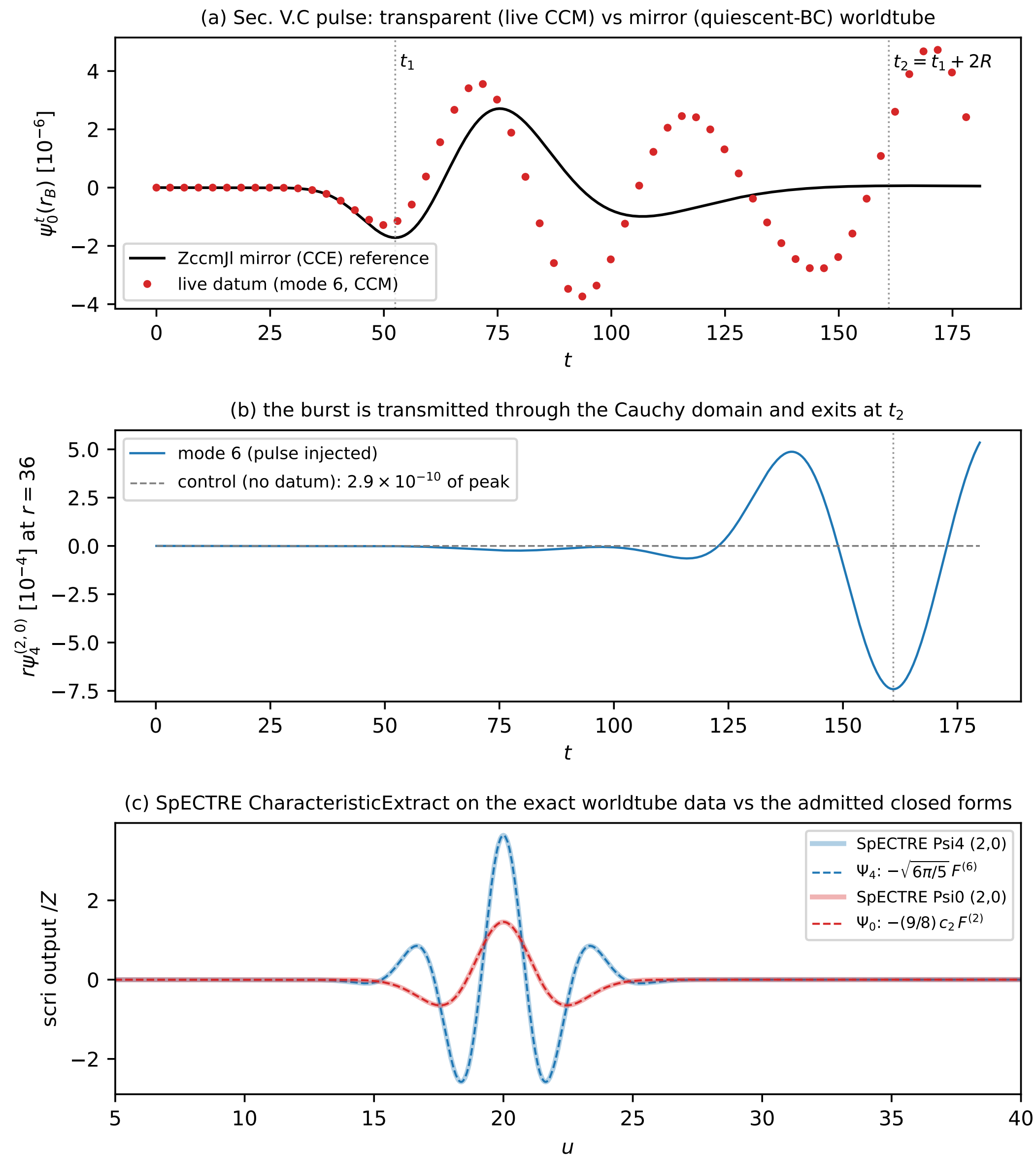

Figure 3. Live-CCM verification (Secs. III C–III D, IV). (a) The characteristic-pulse fidelity test: live ψ_0 datum at the Cauchy boundary vs. the quiescent-worldtube (mirror) reference — early-window agreement 7×10^{-8} of peak; the later $O(1)$ separation is the matching physics. (b) The transmitted burst at the extraction radius exits at $t_2 = t_1 + 2R$; the no-datum control is consistent with zero. (c) SpECTRE `CharacteristicExtract` on the exact worldtube data (41) vs. the closed forms. Generator: `scripts/plot_paper_figures_v2.py`; data: `results/numerical/n14_native/`, `n14_pulse_ref.csv`, `spectre_oracle/`.

figures whose inputs the dataset contains are reproduced bit-faithfully through the authors' own scripts (the remaining published panels lack archived input data and are appendix variants of reproduced main-text figures).

a. Flat Teukolsky, $X = 10^{-5}$ (Fig. 4). Our characteristic transport of the *exact* worldtube data (41) — run through an independent production CCE code [2] at matched worldtube radius 41 and amplitude — agrees with their published scri strain to 1.9×10^{-3} of peak. The deviation is attributable to the integration-tolerance floor of the published runs at this amplitude: tightening our own transport tolerances reduces our deviation from the closed-form strain $\sqrt{6\pi/5}\,F^{(4)}$ by $25\times$ to 4.7×10^{-5}, while the published curves sit at 1.9×10^{-3}. The published CCM and CCE curves themselves differ by only 3.6×10^{-6} of peak — the matching term's effect at $X = 10^{-5}$, the quantitative form of the "perturbatively indistinguishable" statement, peeling-suppressed as in Sec. V B.

b. Characteristic-pulse injection (Fig. 5). For the ingoing-pulse test (Secs. III D, IV D; identical pulse parameters) the boundary ψ_0 first trough matches the published value to 1.4% in amplitude, with a time offset +2.3 in the geometric expectation $[+1.3, +3.0]$ for our worldtube radius 40 versus their 41. The published live (matched) boundary ψ_0 tracks our mirror (quiescent-tube) solver to 1–10% through the main pulse, while our live coupling departs by $O(30\%)$ in the trough window — localizing the $O(\Delta t)$ boundary-holding error of Sec. IV D [O-M5-1].

c. Flat Teukolsky, $X = 2$ (Fig. 6). A dedicated long campaign (eight GPUs) evolves the $X = 2$ system to $t = 80$ at the stable configuration (diss. 0.1, $\kappa_1 = 0.1$, `nghost` $= 4$) with a causally disconnected far-boundary reference (768^3, box ± 192) and boundary-41 Sommerfeld/CCM pairs on a resolution ladder $h = 0.5/0.25/0.125$. At fixed resolution the boundary error (run vs. far-boundary reference) is $E[\mathrm{Somm}] = E[\mathrm{CCM}] = 1.1\times10^{-2}$ with $|\mathrm{CCM}-\mathrm{Somm}| = 8\times10^{-10}$, while the truncation estimate is $E_h = 0.6$: the boundary error sits far below the truncation error and is insensitive to the matching term, reproducing the structure of Ref. [1] at $X = 2$ with a shared linear-initial-data protocol (the constraint-solved-ID difference of the published runs is the documented formulation distinction).

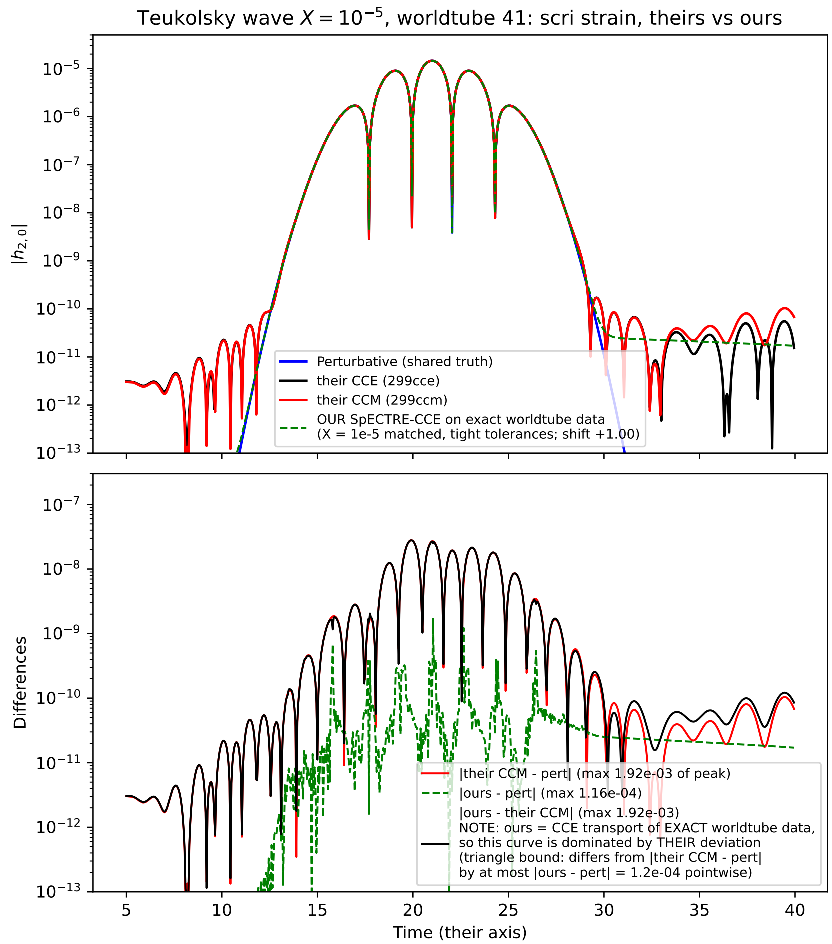

Figure 4. $X = 10^{-5}$ scri strain: published CCM/CCE [1] and our independent characteristic transport of the exact worldtube data (41), against the shared perturbative truth. The lower panel's $|\mathrm{ours} - \mathrm{CCM}|$ tracks $|\mathrm{CCM} - \mathrm{pert}|$ because our transport sits $\sim 40\times$ closer to the truth; the informative curves are the two error budgets. Data: `ref-code/ccm-figures/X=1e-5`, `spectre_oracle`.

VI. CONCLUSIONS

We have formulated Cauchy–characteristic matching for the Z4c system and verified it as a single closed chain (Fig. 1): the ten incoming boundary modes and their data sources (Sec. II); the exact incoming-Weyl dictionary and the Type-II boost carrying the characteristic datum onto the physical rows (Secs. II D–II F); the β-corrected, unringed linear hierarchy that admits live worldtube data in the anchored gauge (Sec. III); the local six-row worldtube map and the retarded matching geometry (Sec. IV); and a verification ladder (Sec. V) running from a native in-process solver, through the verification harness and the exact analytic solution, to an independent production characteristic code and the published reference data. Each link is machine-checked at the precision stated at the corresponding equation.

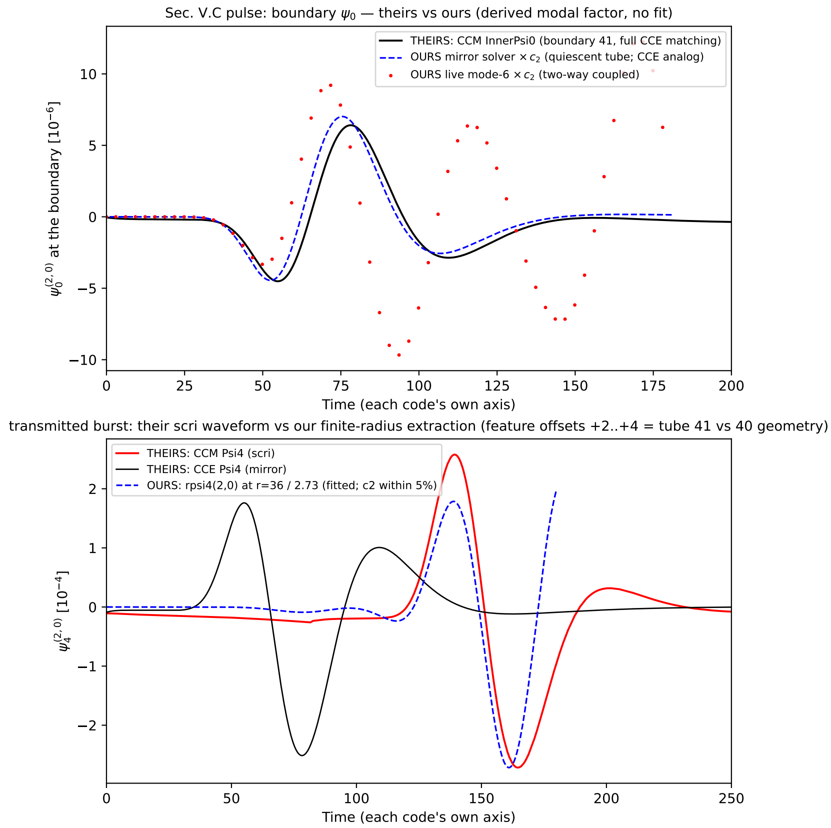

Figure 5. Characteristic-pulse injection: boundary ψ_0 (top) and the transmitted ψ_4 burst (bottom), published data [1] vs. our solver via the derived modal factor. Data: `ref-code/ccm-figures/pulse_on_characteristic_grid`, `n14_native/mode6`.

The principal results are: an exact finite-radius Teukolsky waveform that replaces an auxiliary reference run and converges with measured sixth order (60); a gauge-invariant ψ_0 extraction (38) whose anchored-gauge tails are exactly silent, resolving the representability obstruction of the ringed scheme; and a live, in-process matching loop whose datum fidelity is demonstrated on the characteristic-pulse test and cross-checked, on identical worldtube data, against an independent code to 6×10^{-6} in the waveform and 0.1% in the datum normalization.

The open items are stated where they arise and are collected here. The two-way coupling holds the boundary data fixed across the intra-step substages, an $O(\Delta t)$ approximation whose effect was localized against published data at $O(30\%)$ in the pulse-arrival window [O-M5-1]; linear-in-time boundary interpolation is the corresponding refinement. The strongly nonlinear $X = 2$ test is reproduced here with a shared linear-initial-data protocol, internally consistent but distinct from the constraint-solved initial data of the published runs; a constraint solver is required for amplitude-level $X = 2$ agreement. The black-hole perturbation test of the reference work, with horizon excision and multi-thousand-M evolutions, lies beyond the present resource envelope; its characteristic half is nonetheless exercised by the cross-code verification. Finally, the formulation is developed for the axisymmetric $\ell = 2$ sector at linear order; the higher-multipole and nonlinear hierarchy, the gauge sector of the matching, and the finite-radius–to–scri normalization residual are the natural continuations.

Appendix A: Conventions

Angular operators act in the $q^A \bar{q}_A = 2$ dyad convention; for a spin-weight-s field f, $\eth f = q^A \partial_A f - s\,(q^A \partial_A \ln \rho)\, f$ with the round-sphere connection, and $\bar{\eth}$ its conjugate, so that on the axisymmetric $\ell = 2$ shapes the closures (32) hold. The real shape algebra used throughout is $s \equiv \sin\theta$, $c \equiv \cos\theta$, $P \equiv 3c^2 - 1$, with $\int s^2 \cdot s^2\, d\Omega$, $\int P^2\, d\Omega$, $\int (sc)^2\, d\Omega$ the projection norms of (47); the spin-weighted spherical-harmonic coefficients follow from the conversions (33). The extraction convention for the outgoing waveform is that of the AthenaK Weyl pipeline [6], which equals -1 times the scri member of (56) continued to finite radius; this sign is carried explicitly in (59) and is the convention in which the cross-oracle of Sec. V C returns $\Psi_4 = -\sqrt{6\pi/5}\, F^{(6)}$.

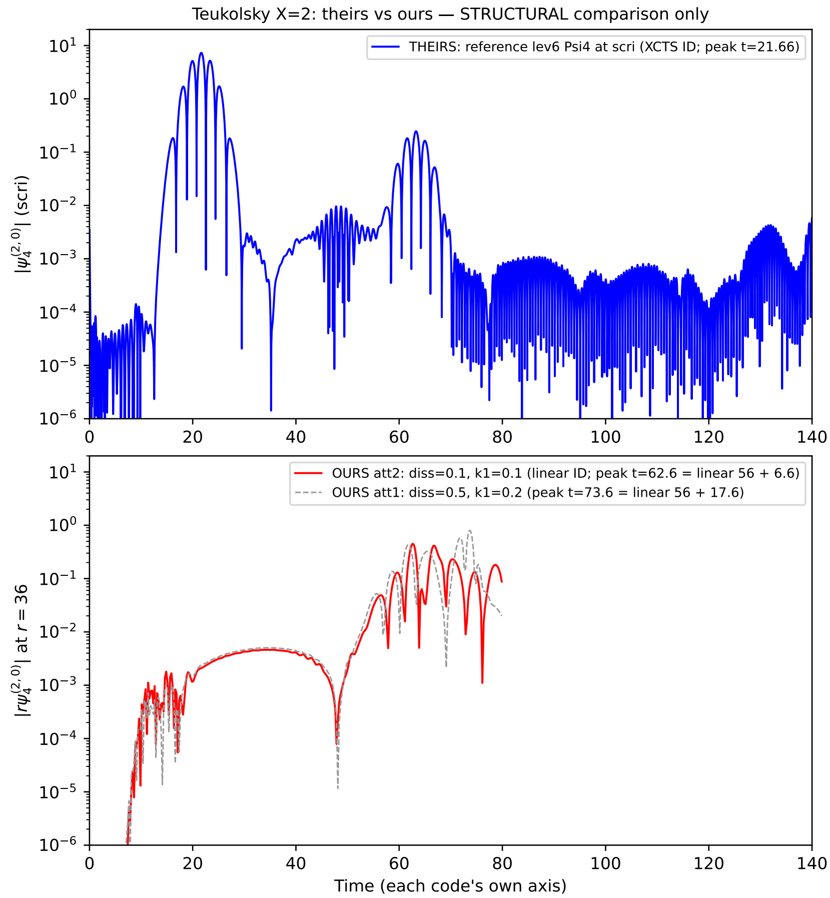

Figure 6. $X = 2$ structural comparison: the published far-boundary reference waveform [1] (top) and our long-campaign boundary-41 runs at two dissipation settings (bottom). The dissipation-dependent peak delay is the linear-initial-data constraint-cleanup signature discussed in Sec. V C. Data: **`results/numerical/ccm_repro_x2_full`**, **`ref-code/ccm-figures/X=2`**.

Appendix B: Exact Teukolsky verification chain

The finite-radius reference (59) and the datum (56) are certified by seven independent symbolic gates (ledger node N12): tracelessness of the metric perturbation (54); the coefficient-exact linearized vacuum equation; the $\sin^2\theta$ factorization of the $(\ell, m) = (2, 0)$ projection; peeling ($k + j = 6$ in the retarded expansion); the scri limit reproducing $-\sqrt{6\pi/5}\,F^{(6)}$; an independent thirty-digit finite-difference cross-check of the Weyl scalar agreeing to 10^{-40}; and the orientation of the magnetic-parity term fixed by the concrete right-handed frame. The Bondi-variable closed forms (41) close the linear hierarchy (34) as a rational identity, and the same forms drive the cross-oracle worldtube file. Two production defects were isolated only by this exact reference: a frame-orientation error in the initial data (the $\hat{e}_\theta$ leg), which violated the linearized vacuum at the 5% level while remaining invisible to constraint monitors, and a one-step offset in the waveform time labels; both are corrected in all data shown.

Appendix C: Constraint-row stability model

The admissibility of the constraint rows 3–6 of Table I is settled on the linearized constraint subsystem at normal incidence (ledger node N13). With $w_\pm = \Theta \pm Z_s$ the characteristic combinations, a frozen-coefficient semidiscrete analysis with the production boundary stencils gives, for every candidate closure (matched advection, the damped sink, a one-sided advected variant, and the characteristic w_- replacement), a maximal real part of the spectrum below 10^{-10} at two resolutions: all are Gustafsson–Kreiss–Sundström stable; the documented nonlinear failure of one realization is attributable to the metric-sector face recomputation, not to the constraint-row concept. A traveling constraint packet is absorbed by the matched-advection realization (which the stock scheme induces) at $1 - 8 \times 10^{-6}$ energy, and by the w_- replacement at $1 - 3 \times 10^{-6}$; the falloff-mismatched sink instead reflects $\sim$10% of the packet.

The matched (stock) realization is therefore the admissible optimum, and the continuum reflection of the matched rows is $|R| = \sigma/(2\omega r)$ with no scheme ambiguity. This is the basis for the production default (ψ_0'-driven physical rows over the stock constraint/gauge rows) quoted in Sec. II C.

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